Spring 2025

Senior Project Report (Final Presentation)

RESCUE: Radar-based Emergency Swarm for Critical Under-rubble Estimation

Group 50:

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Agenda

1. Introduction
2. Problem Statement
3. Project Objectives
4. Design constraints
5. Architecture and implementation
6. Results
7. Avoiding Potential issues
8. Gained knowledge and skills

1 Introduction

utilizes advanced drone swarm technology for search-and-rescue operations.

system integrates radar sensing, cellular communication, and deep learning.

Focused on rapid and accurate detection of humans under rubble.

Addresses challenges in disaster response.

2 Problem Statement

Disaster zones needs quick and accurate survivor detection.

Traditional methods are slow and unreliable, modern methods are needed.

An autonomous system is needed to enhance search-and-rescue efforts, and to deal with harsh conditions.

Project Goals

Objectives

1. Build a group of drones for search and rescue missions.
2. Build customized drones and supply them with the sensors and detecting devices required.
3. Create an autonomous working system (autopilot).
4. Create a server that will be used to monitor and manage the system.

Design Constraints

4 Design Constraints

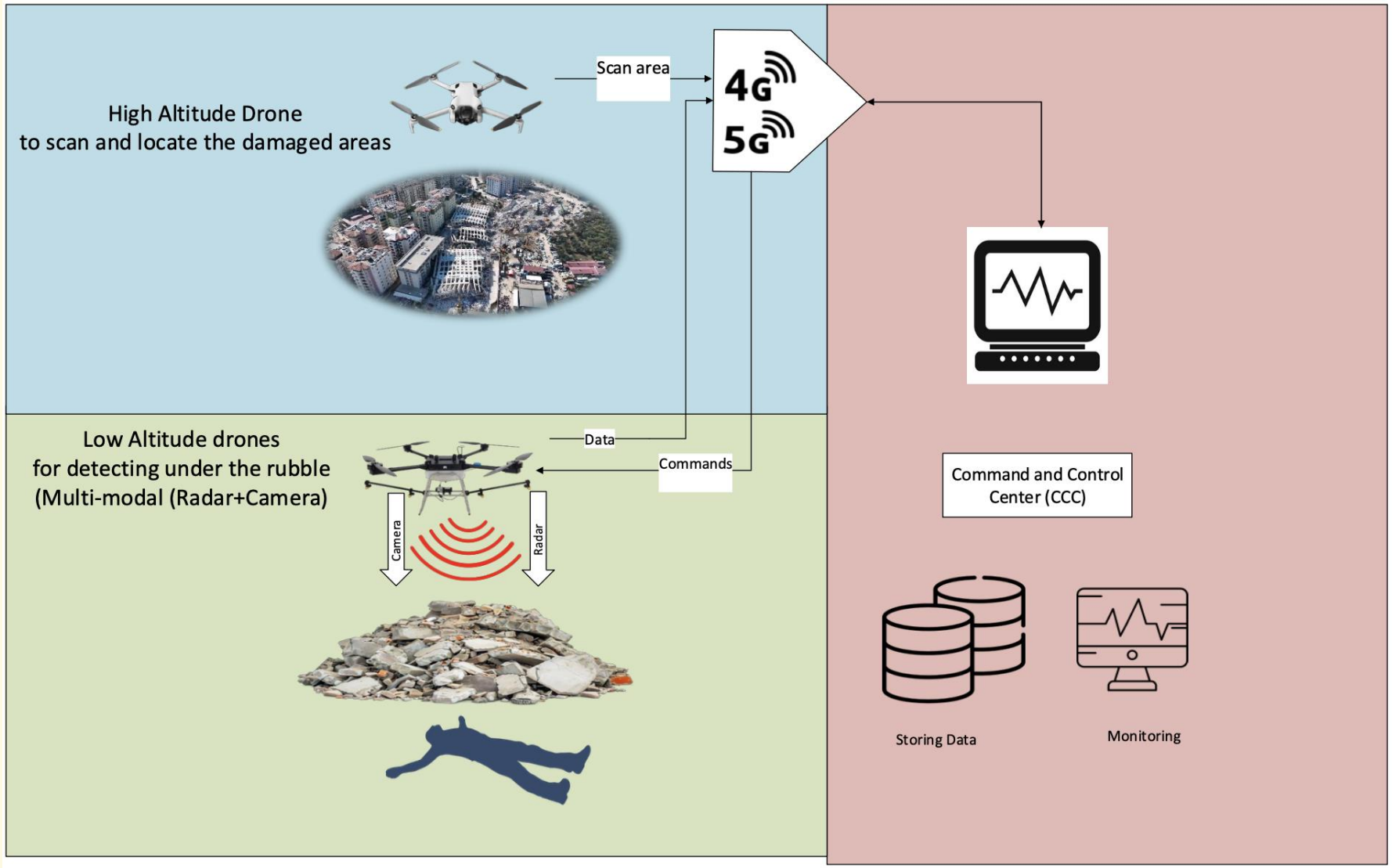
- Battery Life (Flight Duration)
- Thrust-to-Weight Ratio
- Communication Range
- Detection Time for Humans
- Weight
- Positioning Accuracy
- Privacy and Data Security
- Legal Compliance

Architecture and implementation

5 Assumption

Each High-altitude drone scans a separate area to avoid collisions.
There is access to 4/5G base station.
System assumes normal disaster response weather.
It is assumed that the GPS coordinate of the victim/obstacle match the current drone GPS.
Drones communicate only with CCC, not each other.
In LAD edge computing is used to optimize outgoing data traffic.
High-altitude drone flies vertically from center point.

5 High-Level Architecture



5 Main Components of the System

Drone Assembly

Command and Control Center

Communication System

Radar for under-rubble Detection

Camera for Human/Rubble Detection

5 Needed Hardware

	Platform	Justification
Raspberry pi 4	Raspberry pi OS	For Data processing, ML, and managing navigation.
Holybro PX4 Development Kit - X500 v2	Ardupilot firmware	Provides suitable thrust-to-weight ratio and has a suitable size.
Pixhawk 6C flight controller	Ardupilot firmware	Establish full control over drone components
Infineon Radar 24 GHz	Infineon XENSIV SDK	Provides raw data used for precise detection of survivors under the rubble.

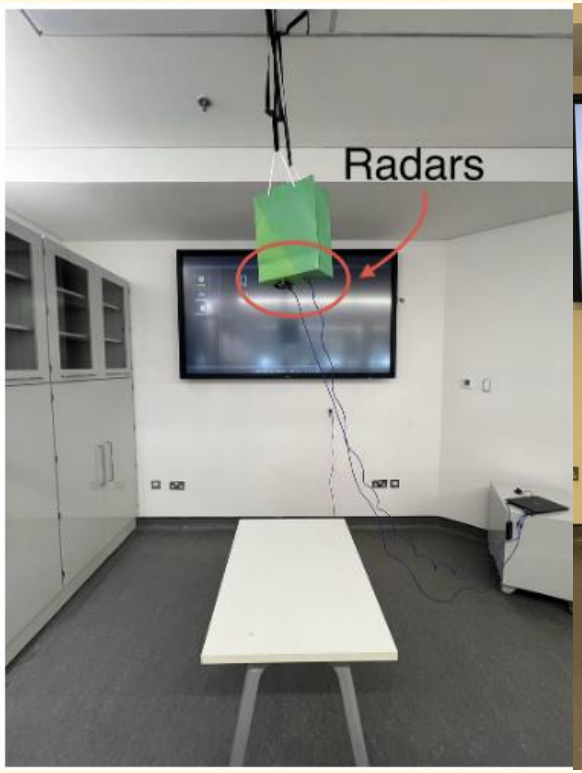
5 Needed Hardware

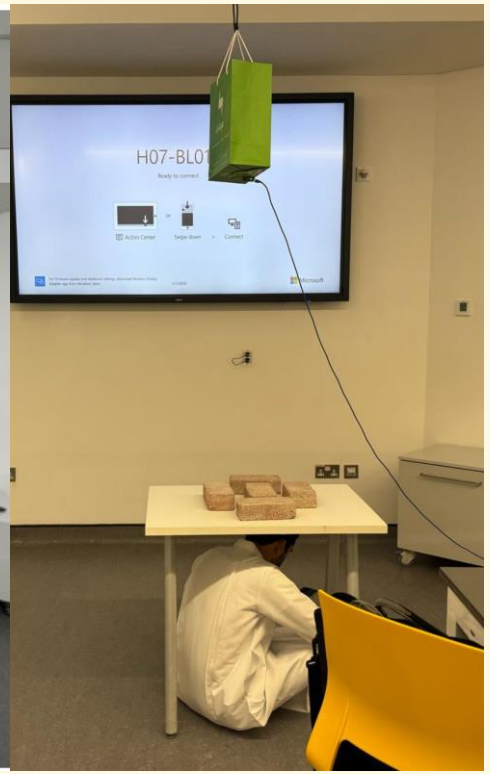
	Platform	Justification
Anker C300 Camera	UVC-compatible (USB video class) works on multiple OS.	for measuring the depth and recognize visual object of interest, can be used for navigation also
D-Link 4G USB modem	D-Link firmware	Used to provide reliable 4g connection, for communication
GAONENG GNB 4S 14.8V 5500mAh LiPo Battery	-	Provides power for the system
RC radio receiver/controller	Pixhawk 6C	For transmitting/receiving controls to the drone from the controller.

5 Needed Software

	platform	Justification
Mission Planner	windows	utilized for controlling the autopilot parameters, establish the flight route in addition to the flight mode, and for controlling the drone using high-level commands.
Google Pub/Sub	Google cloud	for communication management in-between the backend server and the drone swarm.

Data Collection (lab)

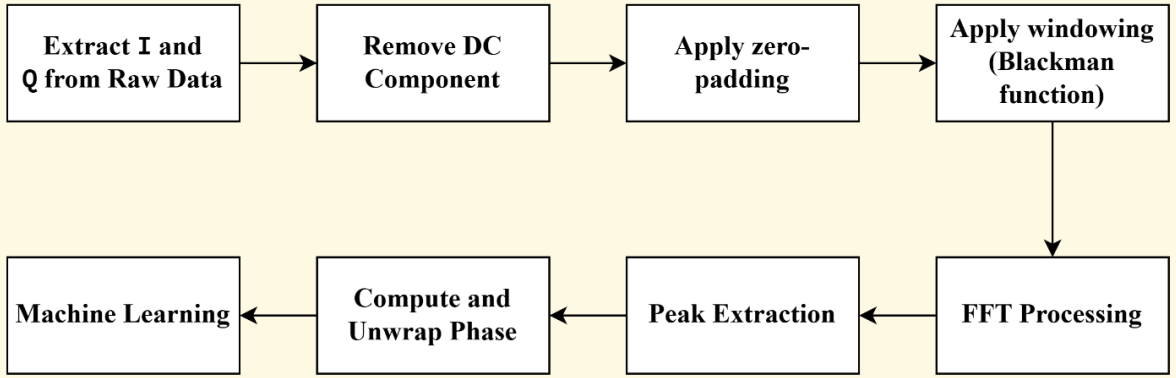


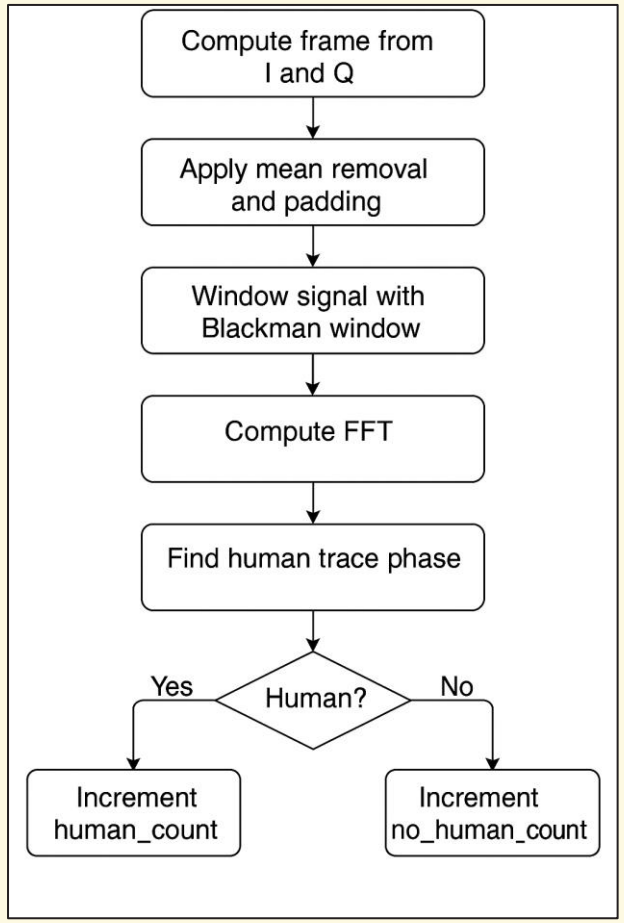


Raw Data was collected from radar sensor, Data is Used to train Deep learning model, Various Deep learning Algorithms were applied

Data Collection (outdoor)

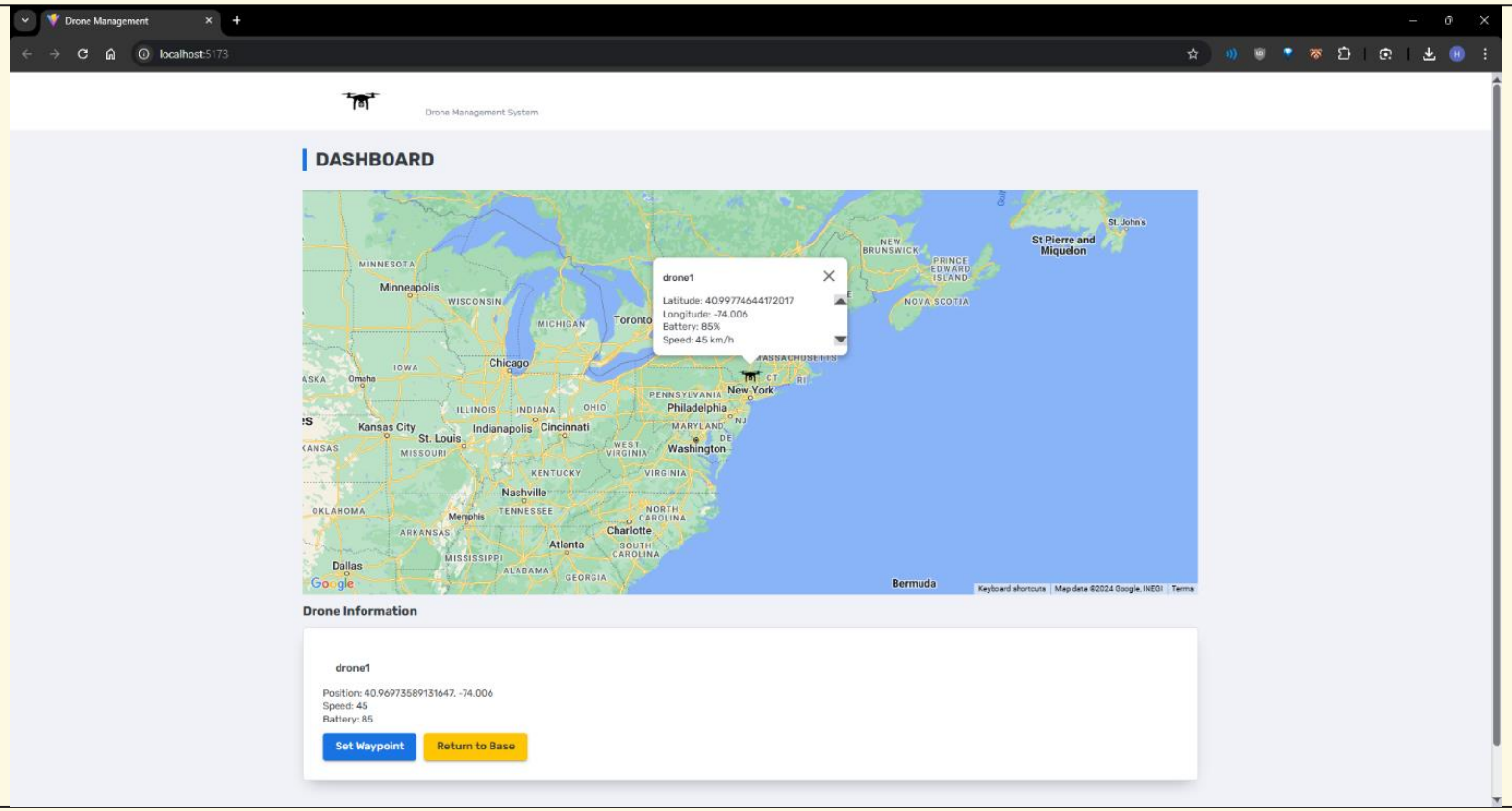
Preprocessing





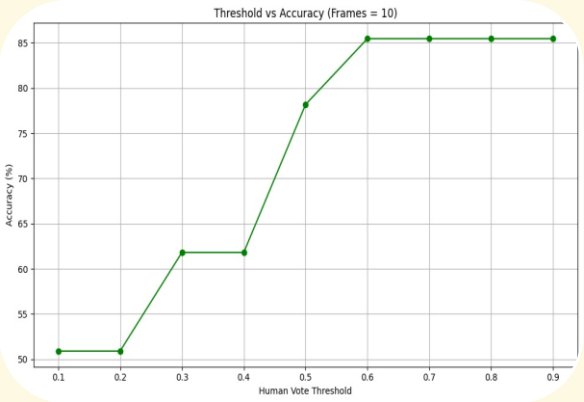
Final Assembled Drone

CCC interface

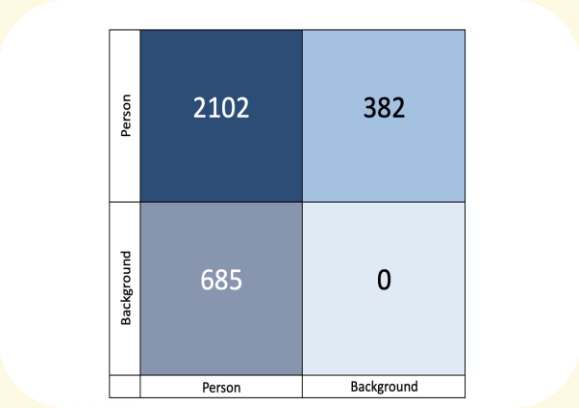


6 Results

Fully Functional Drones with autopilot enabled

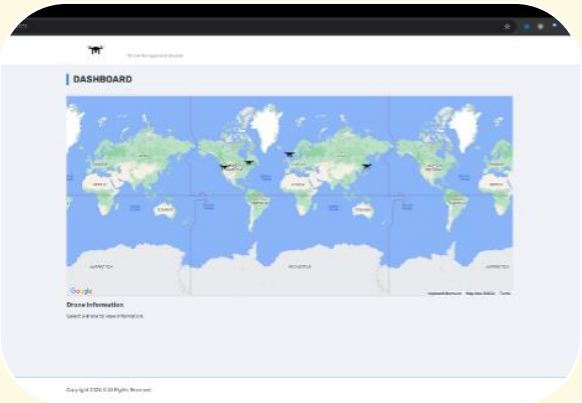


ML model for Radar data with accuracy up to 89.19%

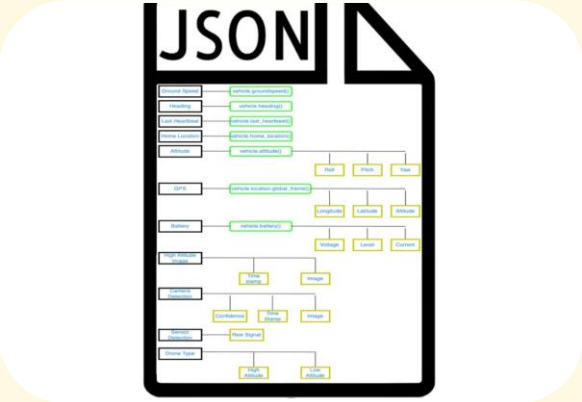


Accurate Vision based model for Human

Accurate Vision based model for Rubble Detection



Functional CCC with a web interface



Efficient communication Protocol between Drone and CCC

6 Results (Testing Constraints)

Constraint	Method of Verification
Battery Life (Flight Duration)	The drone was flown in realistic test conditions, and the flight time was recorded to confirm it exceeds 10 minutes.
Thrust-to-Weight Ratio	All motor thrust values were calculated and compared to the drone's weight to ensure a minimum 2:1 thrust-to-weight ratio, Therefore the drone successfully flew to the required altitude with the calculated thrust.
Weight	The drone payload was weighed including all components (raspberry pi, battery, sensors, camera) to confirm the payload is less than or equal to the 1.5 kg limit.

6 Results (Testing Constraints)

Basic Information

Hovering Time :	13.89 min.	≥ 39.8% Multicopters
Remaining Load :	1.64 kg	≥ 62.1% Multicopters
Max. Takeoff Altitude:	4.78 km	≥ 61.4% Multicopters
Flying Range:	7.54 km	≥ 60.1% Multicopters
Forward Speed:	17.1 m/s	≥ 63.1% Multicopters

Share:

Detail Information

Hovering Performance :

Hovering Time	: 13.89 min.
Throttle Percentage	: 55.4 %
ESC Current	: 4.25 A
Motor Speed	: 6028.9 rpm
Motor Power	: 41.4 W
Battery Voltage	: 16 V
Battery Current	: 19 A
Power Efficiency	: 53.9 %

Max. Throttle Performance :

Flight Time	: 4.2 min.
Total Lift	: 45 N
ESC Current	: 15.7 A
Motor Speed	: 9138.5 rpm
Motor Power	: 144.1 W
Battery Voltage	: 15.7 V
Battery Current	: 62.9 A
Power Efficiency	: 56.7 %

Integral Performance :

Normal Operation	: 11.3 min.
Total Weight	: 2 kg
Remaining Load	: 1.64 kg
Max. Takeoff Altitude	: 4.78 km
Max. Tilt Angle	: 56.7 °
Max. Forward Speed	: 17.1 m/s
Max. Flight Range	: 7.54 km
Wind Resistance	: 5 Degree

6 Results (Testing Constraints)

Constraint	Method of Verification
Detection Time for Humans	We ran detection tests using our trained model on a real dataset, measuring the time taken for each classification; most detections were under 30 seconds.

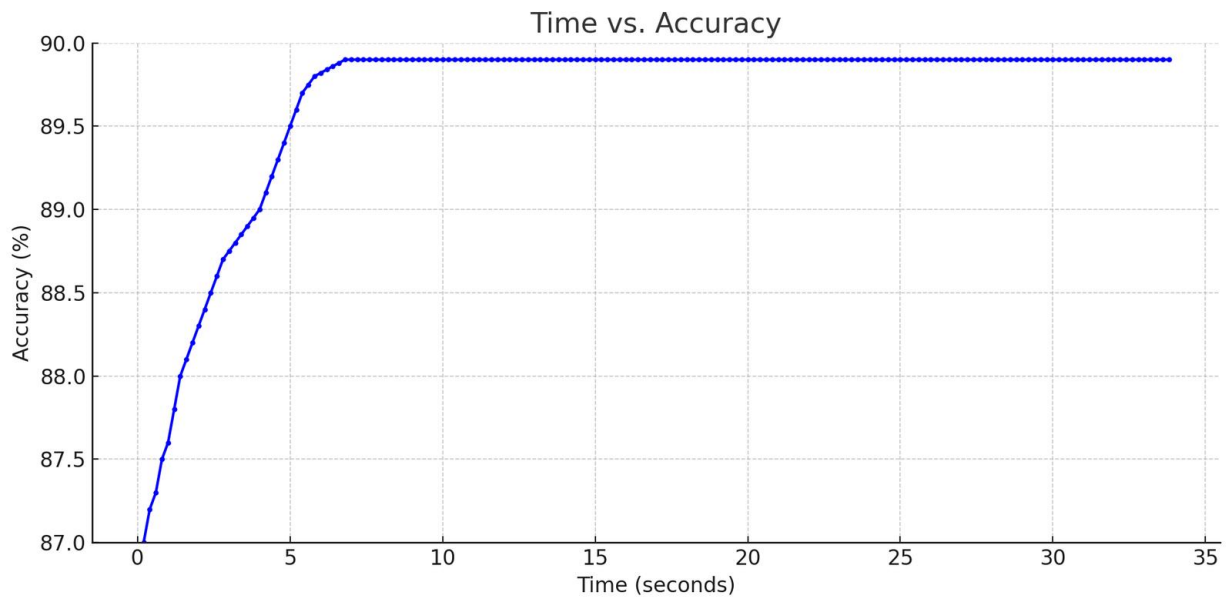
6 Results (Testing Constraints)

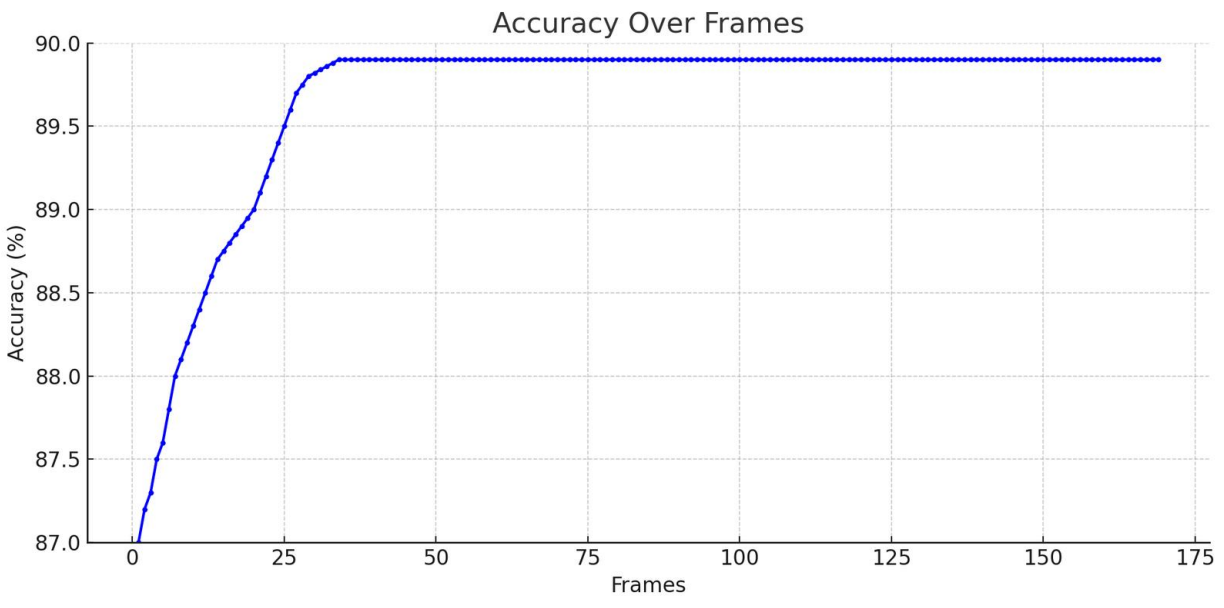
Per 40 frame

Classifier	First Run (s)	Second Run (s)	Third Run (s)	Average (s)	Accuracy
Decision Trees	9.220	9.174	9.190	9.195	78.92%
Random Forest	9.168	9.223	9.235	9.209	88.47%
Gradient Boosting	9.187	9.184	9.217	9.196	89.19%

	First Run (s)	Second Run (S)	Third Run (s)	Average (s)
Camera (YOLO)	3.210	3.198	3.297	3.235

6 Results (Testing Constraints)





7 Avoiding Potential issues

Unpredictable weather	Schedule drone flights for clear, stable weather conditions; have backup test days planned
Limited operational range	Optimize flight paths and monitor real-time power consumption; carry additional batteries for extended missions
Software compatibility issues	Perform early compatibility tests on all software and hardware components; apply standard protocols whenever possible.
Navigation errors	Integrate real-time monitoring and modify navigation algorithms according to environmental data and past flight data.
Risk of drone collision	Send each drone to a different coordinate with. A predefined different path, enforce flight altitude limits, and perform controlled test flights

New Gained Knowledge

- Software Design
- Implementation

New Gained Knowledge

- Choosing and installing drone parts.

- Designing for reliability in tough environment.

- Coordinating high- and low-altitude drones.

New Gained Knowledge

- Drone assembly and calibration.

- Fixed hardware issues like motor and ESC failures.

- Collected data using 24GHz radar sensor and camera

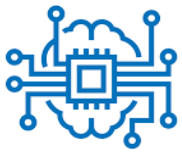
- Using Machine and Deep learning for human detection.

8 Skills Aquired By Group Members

Teamwork and collaboration

Drone Assembly and Configuration

Troubleshooting



Deep learning

Video

Demo

Thank you

Any Questions ?